Exercise 1

To convert a decimal number N to a binary number:

- We divide N by 2 (integer division) and put the remainder of the division aside.
- We divide the previous quotient by 2, and we put the remainder aside again.
- We repeat the division until the quotient is 0.
- The binary number then appears, just take all the remainders from bottom to top.

For N = 14

14 ÷ 2 = 7 ; r : 0
7 ÷ 2 = 3 ; r : 1
3 ÷ 2 = 1 ; r : 1
1 ÷ 2 = 0 ; r : 1

So, 14 (base 10) = 1110 (base 2)

Write JavaScript code that prompts the user for a positive integer and converts that number to binary and displays the result.

Exercise 2

Write the JavaScript code that asks the user to enter a number and checks if it is a perfect number and displays the result.
A number is said to be perfect if it is a non-zero positive integer and if it is equal to the sum of its divisors except itself.
For example: The divisors of 6, except itself, are: 1, 2, and 3.
6 = 1+ 2+3 so 6 is a perfect number.
The perfect ones are rare: 6; 28; 496; 8128; 33 550 336….

Exercise 3

a) Write the JavaScript code to display Figure 1.
b) Modify the previous code to display Figure 2.

Fig. 1

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
```

Fig. 2

```
1
  2
    3
      4
        5
          6
            7
              8
```

Exercise 4

Write the JavaScript code to display the times tables from 1 to 9:

```
1  2  3  4  5  6  7  8  9
2  4  6  8  10 12 14 16 18
3  6  9  12 15 18 21 24 27
4  8  12 16 20 24 28 32 36
5  10 15 20 25 30 35 40 45
6  12 18 24 30 36 42 48 54
7  14 21 28 35 42 49 56 63
8  16 24 32 40 48 56 64 72
9  18 27 36 45 54 63 72 81
```

Exercise 5

Write JavaScript code that allows a user to try to guess the value of a number (between 1 and 100) randomly generated by JavaScript. If the entered value is greater or less, the JavaScript code displays "it's less" or "it's more" respectively until the number is found or the user abandons the search.

If the number is found, it should display "Bravo!!" and the number of attempts made in order to have the correct answer.

The following instruction generates a random number between 1 and 100:

```
let x = Math.floor (( Math.random ()*100)+1);
```

Exercise 6

Write the JavaScript code to display the divisors of each number between 4 and 15 and to indicate whether this number is a prime number.
A prime number is any number divisible only by 1 and itself.
The output should have the following form:

Divisors of 4: 1, 2, 4
Divisors of 5: 1, 5 (Prime)
Divisors of 6: 1, 2, 3, 6
……..